geotech_ORDERING

GEONODE foundation monitoring: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: synthetic data; reuses MITTI-NETRA board. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. No custom circuit board: nothing to order from JLCPCB for this project.
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

3. Bought-in and mechanical items

1. Parts per node

MITTI-NETRA board (see `mitti-netra/`); UART LoRa module breakout (RAK3172 or RA-08 style, AT commands; India 865-867 MHz plan); antenna; protected Li-ion cell; solar panel; ADXL345 and HX711 are on the board; strain gauges and a waterproof potted enclosure. Gateway: any LoRa gateway or a second node plus a Raspberry Pi running `gateway/anomaly.py`.

4. Confidence (read before you order)

- **Order confidence: Medium**: uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet.
- **Works-as-intended confidence: Low-Medium**: Detector logic and feature code are host-tested (54 checks) but on synthetic vibration; the crack signature is an assumption; node hardware is the MITTI-NETRA board with a LoRa module.
- **First thing to do:** Record real vibration for a week and re-derive the thresholds before trusting any alarm.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.